

The relative position of lines in space

Parallelism revisited where a third case exists — and the two theorems that make it usable.

LESSON 19–20

2 × 40 MIN

GEOMETRY 10, §9

EXTENSION

LESSON CLOCK

10'

20'

22'

20'

16'

2'

The definition, restated	10 min
One parallel through a point	20 min
Transitivity	22 min
Respectively parallel sides	20 min
Practice	16 min
Homework	2 min

LEARNING OBJECTIVES

- State the definition of parallel lines in space.
- Prove that through a point not on a line there is exactly one parallel to it.
- Apply the transitivity of parallelism in space.
- Prove the theorem on the angle between lines with respectively parallel sides.

TEXTBOOK

National	pp. 83–92
Cambridge	Core & Extended, pp. 277–281
Workbook	Exercise 9.1

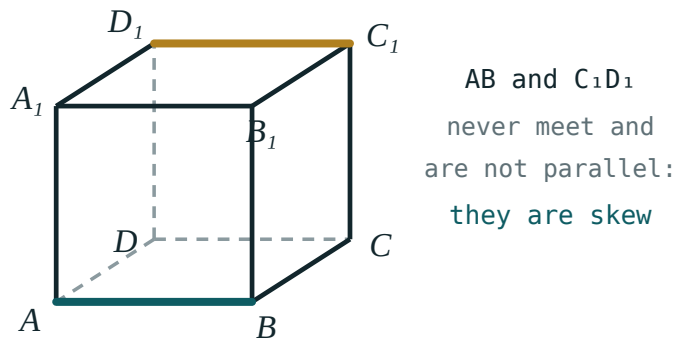
01

Explanation

The definition needs one more word

PARALLEL IN SPACE

Two lines are **parallel** when they lie in **one plane** and do not meet. Without the first condition the definition would also admit skew lines, which never meet either.



*Skew lines never meet — and no plane holds both.
That is exactly what parallelism excludes.*

Everything else about parallel lines carries over from the plane unchanged. What must be re-proved is that they still **exist and are unique** once the extra dimension is available.

One parallel through a point

Theorem. Through a point M not on a line a there passes exactly one line parallel to a .

Proof. The line a and the point M determine a unique plane α (Theorem 1 of Lesson 4–5). Inside α the plane axiom gives exactly one parallel through M . Any parallel to a through M must be coplanar with a by the definition, and that plane contains a and M , so it is α . Hence there is no other. ■

THE PATTERN OF EVERY PROOF IN THIS CHAPTER

Reduce the space problem to a plane problem by naming the right plane, then use plane geometry. Naming the plane is the whole of the work.

Transitivity

Theorem. If $a \parallel b$ and $b \parallel c$ then $a \parallel c$.

In the plane this is obvious. In space it is not, because a and c need not obviously share a plane — and until that is proved, they might be skew. The proof constructs the plane and shows they cannot meet.

PERPENDICULARITY IS NOT TRANSITIVE

In space, $a \perp b$ and $b \perp c$ does **not** give $a \perp c$. Take three edges of a cube meeting at one vertex: any two are perpendicular, so the conclusion would be false. Parallelism is transitive; perpendicularity

is not.

Respectively parallel sides

Theorem. If two angles in space have respectively parallel sides, they are either equal or supplementary.

This is what allows the angle between skew lines to be well defined: translating a line does not change the angle it makes with anything, because the translated line is parallel to the original and its sides are respectively parallel.

translate b to meet $a \implies$ the angle is unchanged

Which of “equal” or “supplementary” occurs depends on the directions chosen; the convention is to take the acute one.

02 Worked examples

EXAMPLE 1 In cube $ABCD A_1 B_1 C_1 D_1$, prove $AB \parallel D_1 C_1$.

① $AB \parallel DC$
Opposite sides of a square face.

② $DC \parallel D_1 C_1$
Opposite sides of face $DCC_1 D_1$.

③ By transitivity.

ANSWER

$AB \parallel D_1 C_1$

EXAMPLE 2 Give three lines in a cube with $a \perp b$ and $b \perp c$ but a not perpendicular to c .

- ① Take $a = AB$, $b = AA_1$, $c = D_1C_1$.
- ② $AB \perp AA_1$
Adjacent edges of a face.
- ③ $AA_1 \perp D_1C_1$
A vertical edge against a horizontal one.
- ④ But $AB \parallel D_1C_1$, so the angle between them is 0° .
Not perpendicular.

ANSWER

AB, AA_1, D_1C_1 — the first and last are parallel

EXAMPLE 3 M is a point not on line a . How many lines through M are skew to a ?

- ① Only one is parallel.
- ② Those in the plane α meet a or are that parallel.
- ③ Every other line through M leaves α .

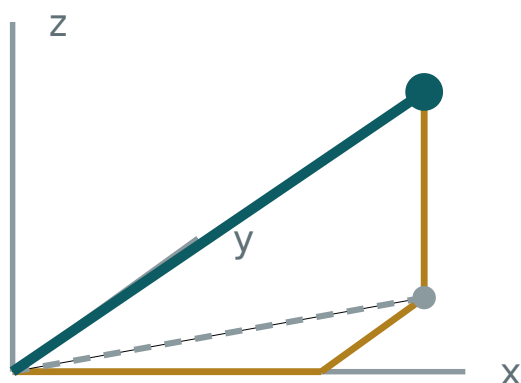
ANSWER

Infinitely many

03 Interactive model

Hold two pencils parallel, then move one out of the plane to show what the definition excludes.

Lines through a point



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Parallel lines in space	Fazoda parallel chiziqlar	Параллельные прямые в пространстве
Coplanar lines	Bir tekislikdagi chiziqlar	Компланарные прямые
Transitivity	Tranzitivlik	Транзитивность
Respectively parallel sides	Mos parallel tomonlar	Соответственно параллельные стороны
Unique parallel	Yagona parallel	Единственная параллельная
Direction of a line	Chiziq yo'nalishi	Направление прямой
Angle between lines	Chiziqlar orasidagi burchak	Угол между прямыми
Auxiliary plane	Yordamchi tekislik	Вспомогательная плоскость

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Parallel lines in space must:

not meet

lie in one plane and not meet

be equal

be perpendicular to a third

Through a point off a line the number of parallels is:

0

1

2

infinitely many

Parallelism in space is:

not transitive

transitive

symmetric only

undefined

Perpendicularity in space is:

transitive

not transitive

the same as parallelism

undefined

Angles with respectively parallel sides are:

always equal

equal or supplementary

always supplementary

unrelated

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Define parallel lines in space

ANSWER coplanar and never meeting

2. How many parallels through a point off a line?

ANSWER 1

3. Is $AB \parallel D_1C_1$ in a cube?

ANSWER yes

4. Is parallelism transitive in space?

ANSWER yes

5. Is perpendicularity transitive in space?

ANSWER no

6. Name a line skew to AB in a cube

ANSWER CC_1 (or DD_1)

7. Two lines that never meet are parallel?

ANSWER not necessarily — they may be skew

Medium · the standard question

7 PROBLEMS

1. Prove $AB \parallel D_1C_1$ in a cube

ANSWER Transitivity through DC

2. How many lines through a point are skew to a given line?

ANSWER infinitely many

3. If $a \parallel b$ and b meets c , what can a and c be?

ANSWER meeting or skew

4. Angle between AB and C_1D_1 in a cube

ANSWER 0° — they are parallel

5. Angle between AB and B_1C_1

ANSWER 90°

6. Give two lines perpendicular to the same line but not parallel

ANSWER Two edges of a cube meeting a third at one vertex

7. Why is the angle between skew lines well defined?

ANSWER Respectively parallel sides give the same angle

Hard · stretch and reason

7 PROBLEMS

1. Prove transitivity of parallelism in space

ANSWER Construct the plane through a and a point of c , then argue in it

2. Prove: if $a \parallel b$ and a plane meets a , it meets b or contains it

ANSWER Otherwise $b \parallel \alpha$ forces $a \parallel \alpha$

3. In a cube, count the pairs of parallel edges

ANSWER 18 — three directions, $C(4,2) = 6$ each

4. In a cube, count the pairs of intersecting edges

ANSWER 24 — $C(3,2) = 3$ at each of 8 vertices

5. In a cube, count the pairs of skew edges

ANSWER 24

6. Check the three counts against $C(12,2)$

ANSWER $18 + 24 + 24 = 66 \checkmark$

7. Prove the theorem on respectively parallel sides

ANSWER Translate one angle onto the other; the triangles are congruent

07 Homework

Homework — 4 tasks

Every proof must name the plane it works in.

1. Prove that through a point not on a line there is exactly one parallel to it.
2. In cube $ABCD A_1 B_1 C_1 D_1$, prove $AD \parallel B_1 C_1$.
3. Give a counterexample showing perpendicularity in space is not transitive.
4. Explain in three sentences why the definition of parallel lines needs the words “in one plane”.

Skew lines

The case the plane does not have — its test, its angle, and the common perpendicular that measures the distance between two lines that never meet.

LESSON 21–22

2 × 40 MIN

GEOMETRY 10, §10

EXTENSION

LESSON CLOCK

10'

22'

24'

20'

20'

4

The test	10 min
The angle, by translation	22 min
The common perpendicular	24 min
Distances in a cuboid	20 min
Practice	20 min
Homework	4 min

LEARNING OBJECTIVES

- State and apply the test for skew lines.
- Find the angle between two skew lines by translation.
- Describe the common perpendicular of two skew lines.
- Compute the distance between skew edges of a cuboid.

TEXTBOOK

National	pp. 93–102
Cambridge	Core & Extended, pp. 277–281
Workbook	Exercise 10.1

01

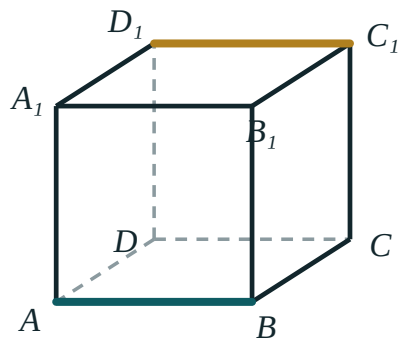
Explanation

The test

TEST FOR SKEW LINES

If one line lies in a plane and the other meets that plane at a point **not on** the first line, then the two lines are skew.

This is the practical test. Checking the definition directly — “no plane contains both” — requires ruling out infinitely many planes; the test requires exhibiting one.



AB and C_1D_1
never meet and
are not parallel:
they are skew

AB lies in the base; CC_1 meets the base at C, which is not on AB. Skew.

In cube $ABCD A_1 B_1 C_1 D_1$: AB lies in the plane $ABCD$, and CC_1 meets that plane at C , which is not on AB . So AB and CC_1 are skew.

The angle

Translate either line until it meets the other, and measure the angle between the resulting intersecting lines. By the theorem on respectively parallel sides, the answer does not depend on which line is moved or where it is moved to.

$$\text{angle}(a, b) = \text{angle}(a', b) \text{ whenever } a' \parallel a$$

In a cube, AB and B_1C_1 : translate B_1C_1 down to BC . Since $AB \perp BC$, the angle is 90° .
Two lines can be perpendicular without ever meeting.

TRANSLATE, DO NOT ROTATE

Only a translation preserves direction. Sliding a line along itself, or turning it, changes the answer.

The common perpendicular

THEOREM

Two skew lines have exactly one **common perpendicular** — a segment meeting both and perpendicular to both. Its length is the **distance** between the lines, and it is the shortest segment joining a point of one to a point of the other.

In a cube of side a , the common perpendicular of AB and CC_1 is BC , of length a . It meets AB at B and CC_1 at C , and is perpendicular to both.

A second method. Take the plane through one line parallel to the other. The distance between the skew lines equals the distance from the second line to that plane — a construction that generalises to cases where the common perpendicular is hard to see.

Distances in a cuboid

PAIR IN CUBOID $A \times B \times C$	COMMON PERPENDICULAR	DISTANCE
AB and CC_1	BC	b
AB and DD_1	AD	b
AA_1 and BC	AB	a
AB and D_1C_1	they are parallel	—

In a cube every edge has four skew partners, and in each case the common perpendicular is another edge. The picture is easier than the algebra, which is why the diagram is drawn first.

02 Worked examples

EXAMPLE 1 In cube $ABCD A_1 B_1 C_1 D_1$ of side 6, find the distance between AB and CC_1 .

1

BC meets AB at B and CC_1 at C .

2

$BC \perp AB$ and $BC \perp CC_1$.
It is the common perpendicular.

3

$|BC| = 6$

ANSWER

6

EXAMPLE 2 Are AD and B_1C_1 skew in a cube?

- ① $B_1C_1 \parallel BC$
Opposite sides of face BCC_1B_1 .
- ② $BC \parallel AD$
Opposite sides of the base.
- ③ By transitivity $B_1C_1 \parallel AD$.
- ④ Parallel lines are coplanar, so they are not skew.
Always test for parallelism first.

ANSWER

No — they are parallel

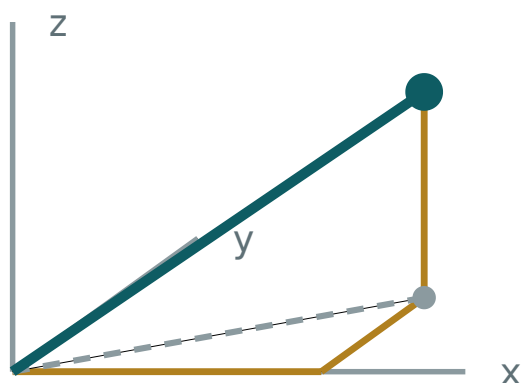
EXAMPLE 3 Cuboid $4 \times 6 \times 8$. Find the angle between AB and DB_1 .

- ① $AB = 4$ along x ; DB_1 from $D(0,6,0)$ to $B_1(4,0,8)$.
- ② Translate AB to DC , which meets DB_1 at D .
- ③ $DC = 4$, $DB_1 = \sqrt{16 + 36 + 64} = \sqrt{116}$
- ④ $\cos \theta = \frac{4}{\sqrt{116}} \approx 0.3714$

ANSWER $\approx 68.2^\circ$ **03 Interactive model**

Hold two pencils skew, then slide one parallel until they cross — the angle is unchanged.

Skew edges of a box



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Skew lines	Ayqash to'g'ri chiziqlar	Скрещивающиеся прямые
Test for skew lines	Ayqashlik alomati	Признак скрещивающихся прямых
Common perpendicular	Umumiy perpendikulyar	Общий перпендикуляр
Distance between skew lines	Ayqash chiziqlar orasidagi masofa	Расстояние между скрещивающимися прямыми
Angle between skew lines	Ayqash chiziqlar burchagi	Угол между скрещивающимися прямыми
Translation of a line	Chiziqni ko'chirish	Перенос прямой
Shortest segment	Eng qisqa kesma	Кратчайший отрезок
Non-coplanar	Bir tekislikda yotmaydigan	Не лежащие в одной плоскости

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The test for skew lines needs:

two planes

one line in a plane and the other meeting it off that line

a common point

equal lengths

Two skew lines have how many common perpendiculars?

0

1

2

infinitely many

The distance between skew lines is:

zero

the common perpendicular's length

infinite

undefined

To find the angle between skew lines you:

rotate one

translate one

project both

measure directly

Skew lines can be perpendicular:

never

always

sometimes

only in a cube

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. State the test for skew lines

ANSWER One line in a plane, the other meeting it off that line

2. Are AB and CC_1 skew in a cube?

ANSWER yes

3. Are AB and D_1C_1 skew?

ANSWER no — parallel

4. Common perpendicular of AB and CC_1

ANSWER BC

5. Distance between them in a cube of side 5

ANSWER 5

6. Angle between AB and B_1C_1

ANSWER 90°

7. How many edges are skew to a given edge of a cube?

ANSWER 4

Medium · the standard question

7 PROBLEMS

1. Cube of side 6: distance between AB and CC_1

ANSWER 6

2. Cube of side 6: distance between AA_1 and BC

ANSWER 6

3. Angle between AB and DC_1 in a cube

ANSWER 45°

4. Angle between AB_1 and BC_1 in a cube

ANSWER 60°

5. Cuboid $4 \times 6 \times 8$: angle between AB and DB_1

ANSWER $\approx 68.2^\circ$

6. Common perpendicular of AA_1 and CD in a cube

ANSWER AD

7. List the four edges skew to AD

ANSWER $BB_1, CC_1, A_1B_1, D_1C_1$

Hard · stretch and reason

7 PROBLEMS

1. Cube of side a : distance between AB and B_1D_1

ANSWER $\frac{a}{\sqrt{2}} \approx 0.707a$

2. Cube of side a : distance between the diagonals AC and B_1D_1

ANSWER a

3. Cube of side a : distance between AB and A_1C_1

ANSWER a

4. Prove the common perpendicular is unique

ANSWER Two would force the lines to be coplanar

5. Prove the common perpendicular is the shortest joining segment

ANSWER Any other joins via a right triangle with it as a leg

6. Cuboid $3 \times 4 \times 12$: distance between AB and CC_1

ANSWER 4

7. Explain the plane method for the distance between skew lines

ANSWER Take the plane through one parallel to the other; the distance is from the second to it

07 Homework

Homework — 5 tasks

Draw the cube once, large, and answer tasks 1–3 from it.

1. In cube $ABCD A_1 B_1 C_1 D_1$, list every edge skew to CD .
2. Find the common perpendicular and the distance between AD and CC_1 for side 8.
3. Find the angle between AD and BC_1 .
4. Cuboid $5 \times 12 \times 9$: find the angle between AB and DB_1 .

5. Explain in three sentences why the common perpendicular is the shortest segment joining the two lines.

The relative position of a line and a plane

Three cases, one test — and the theorem that turns a hard space question into an easy plane one.

LESSON 23–24

2 × 40 MIN

GEOMETRY 10, §11

EXTENSION

LESSON CLOCK

8'	22'	24'	22'	18'	6'
The three cases			8 min		
The test			22 min		
The converse theorem			24 min		
Applications in a cube			22 min		
Practice			18 min		
Homework			6 min		

LEARNING OBJECTIVES

- List the three positions of a line and a plane.
- State and apply the test for a line parallel to a plane.
- Prove that a plane through a line parallel to another meets it in a parallel line.
- Use the results in a cube or prism.

TEXTBOOK

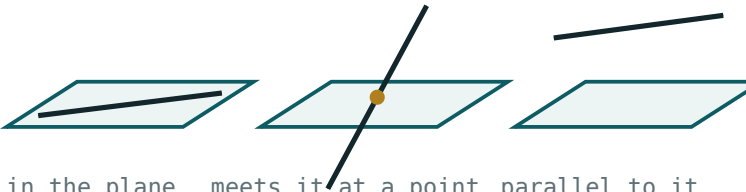
National	pp. 103–112
Cambridge	Core & Extended, pp. 277–281
Workbook	Exercise 11.1

01

Explanation

The three cases

a line and a plane



in the plane meets it at a point parallel to it

In the plane, meeting it at a point, parallel to it. There is no fourth case.

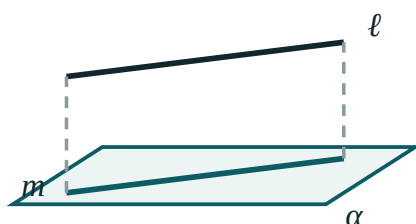
CASE	COMMON POINTS	NOTATION
the line lies in the plane	infinitely many	$a \subset \alpha$
the line meets the plane	1	$a \cap \alpha = M$
the line is parallel to the plane	0	$a \parallel \alpha$

Two common points would put the whole line in the plane, by axiom S2 — which is why there is no case with exactly two.

The test

TEST FOR A LINE PARALLEL TO A PLANE

If a line ℓ not lying in a plane α is parallel to **some** line m lying in α , then $\ell \parallel \alpha$.



$\ell \parallel m$ and m lies in α , so $\ell \parallel \alpha$

ℓ is parallel to m , and m lies in α . That is enough.

Proof. ℓ and m are parallel, so they lie in one plane β . If ℓ met α , the meeting point would lie in $\beta \cap \alpha = m$ — so it would be on m , contradicting $\ell \parallel m$. ■

“SOME”, NOT “EVERY”

You need to exhibit **one** line of the plane parallel to ℓ . Finding it is the whole exercise; in a cube it is almost always the opposite edge of a face.

The converse

Theorem. If $\ell \parallel \alpha$ and a plane β contains ℓ and meets α in a line m , then $\ell \parallel m$.

Proof. ℓ and m both lie in β , so they are coplanar. If they met, the meeting point would be on ℓ and in α — impossible, since $\ell \parallel \alpha$. ■

THIS IS THE SECTION RULE R2, PROVED

It is why a cutting plane meets two parallel faces in parallel lines: each face is parallel to the other, and the cutting plane contains a line of each.

In a cube

In $ABCD A_1 B_1 C_1 D_1$, is $A_1 B_1$ parallel to the plane $ABCD$?

1. $A_1 B_1$ does not lie in $ABCD$.
2. $A_1 B_1 \parallel AB$, opposite sides of the square face $ABB_1 A_1$.
3. $AB \subset ABCD$.
4. By the test, $A_1 B_1 \parallel ABCD$. ■

The **distance** from a line parallel to a plane to that plane is the same at every point — here it is the height of the cube. That constancy is what makes the distance well defined.

02 Worked examples

EXAMPLE 1 Prove that $B_1 C_1$ is parallel to the plane $ABCD$ in a cube.

① $B_1 C_1 \not\subset ABCD$

It is on the top face.

② $B_1 C_1 \parallel BC$

Opposite sides of face $BCC_1 B_1$.

③ $BC \subset ABCD$

④ Test applies.

ANSWER

$B_1 C_1 \parallel ABCD$

EXAMPLE 2 A plane through A_1B_1 meets the base $ABCD$ in a line m . What can be said about m ?

① $A_1B_1 \parallel ABCD$

By the previous example.

② The cutting plane contains A_1B_1 and meets the base in m .

③ By the converse theorem, $m \parallel A_1B_1$.

ANSWER

$m \parallel A_1B_1$, so also $m \parallel AB$

EXAMPLE 3 Is the diagonal AC_1 of a cube parallel to any face?

① It meets the base at A and the top at C_1 .

② So it is not parallel to those two.

③ It meets each of the other four faces too.
It is a space diagonal.

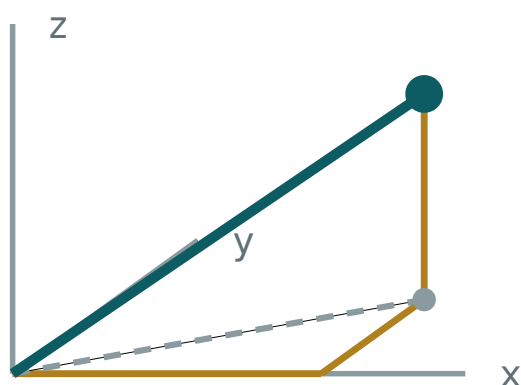
ANSWER

No — a space diagonal meets every face

03 Interactive model

Hold a pencil parallel to the table and ask which line on the table it is parallel to.

Line and plane



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

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Line parallel to a plane	Tekislikka parallel chiziq	Прямая, параллельная плоскости
Test for parallelism	Parallellik alomati	Признак параллельности
Line lying in a plane	Tekislikda yotuvchi chiziq	Прямая, лежащая в плоскости
Point of intersection	Kesishish nuqtasi	Точка пересечения
Auxiliary plane	Yordamchi tekislik	Вспомогательная плоскость
Trace	Iz	След
Distance from a line to a plane	Chiziqdan tekislikkacha masofa	Расстояние от прямой до плоскости
Section	Kesim	Сечение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself

A line and a plane can share:

The test for $\ell \parallel \alpha$ needs:

If $\ell \parallel \alpha$ and $\beta \supset \ell$ cuts α in m , then:

A_1B_1 and the plane $ABCD$ in a cube are:

meeting

parallel

coincident

perpendicular

A space diagonal of a cube is parallel to:

the base

a side face

no face

every face

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Name the three positions of a line and a plane

ANSWER in it, meeting it, parallel

2. Can a line and a plane share exactly 2 points?

ANSWER no

3. State the test for $\ell \parallel \alpha$

ANSWER $\ell \parallel m$ for some $m \subset \alpha$

4. Is $A_1B_1 \parallel ABCD$ in a cube?

ANSWER yes

5. Is $AA_1 \parallel ABCD$?

ANSWER no — it meets it at A

6. Is AC_1 parallel to any face?

ANSWER no

7. Distance from a parallel line to a plane is:

ANSWER constant

Medium · the standard question

7 PROBLEMS

1. Prove $B_1C_1 \parallel ABCD$

ANSWER It is parallel to BC , which lies in the base

2. Prove $A_1C_1 \parallel ABCD$

ANSWER $A_1C_1 \parallel AC \subset ABCD$

3. Which faces is AB parallel to?

ANSWER $A_1B_1C_1D_1$ and DCC_1D_1

4. Which faces does AB lie in?

ANSWER $ABCD$ and ABB_1A_1

5. Which faces does AB meet at a point?

ANSWER ADD_1A_1 and BCC_1B_1

6. A plane through AB meets the top face in m . Then m

ANSWER $\parallel AB$

7. Distance from A_1B_1 to the base in a cube of side 7

ANSWER 7

Hard · stretch and reason

7 PROBLEMS

1. Prove the test for a line parallel to a plane

ANSWER A meeting point would lie on the line of intersection

2. Prove the converse theorem

ANSWER ℓ and m are coplanar in β and cannot meet

3. Through a point off a plane, how many lines are parallel to the plane?

ANSWER infinitely many — a whole plane of them

4. Prove: if $\ell \parallel \alpha$ and $\ell \parallel \beta$ with $\alpha \cap \beta = m$, then $\ell \parallel m$

ANSWER Apply the converse in each plane

5. In a cube, find a plane parallel to AC but not containing it

ANSWER $A_1B_1C_1D_1$

6. Give a line parallel to two intersecting planes

ANSWER One parallel to their line of intersection

7. Why can a line parallel to a plane never meet a line of that plane?

ANSWER The meeting point would be common to line and plane

07 Homework

Homework — 5 tasks

Each proof must name the line of the plane it uses.

1. Prove that D_1C_1 is parallel to the plane $ABCD$.
2. List every face of a cube that AD is parallel to.
3. A plane through B_1C_1 meets the base in a line m . What is m parallel to, and why?
4. Prove that a line parallel to two intersecting planes is parallel to their line of intersection.
5. Explain in three sentences why a line and a plane can never share exactly two points.

The relative position of planes in space

Parallel planes, the test that establishes them, and the four properties that follow immediately.

LESSON 25–26

2 × 40 MIN

GEOMETRY 10, §12

EXTENSION

LESSON CLOCK

8'

22'

26'

20'

18'

6'

The three cases	8 min
The test	22 min
Four properties	26 min
Applications	20 min
Practice	18 min
Homework	6 min

LEARNING OBJECTIVES

- List the three positions of two planes.
- State and apply the test for parallel planes.
- Prove that a third plane cuts two parallel planes in parallel lines.
- Use parallel planes to prove segments equal.

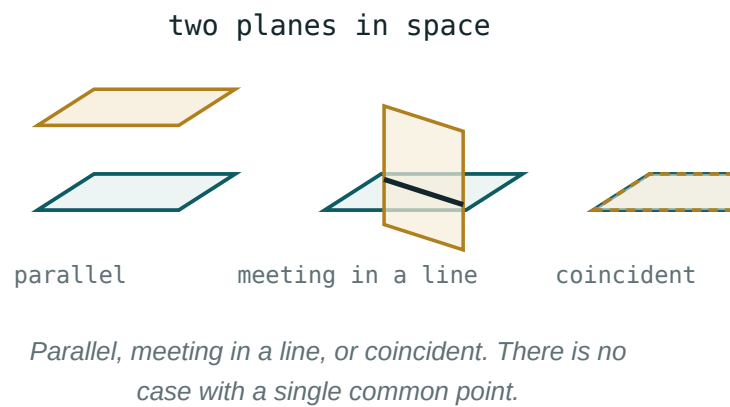
TEXTBOOK

National	pp. 113–124
Cambridge	Core & Extended, pp. 277–281
Workbook	Exercise 12.1

01

Explanation

The three cases



Two planes with a common point share a whole line, by axiom S3. So either they have no common point (parallel), a common line (intersecting), or they are the same plane.

The test

TEST FOR PARALLEL PLANES

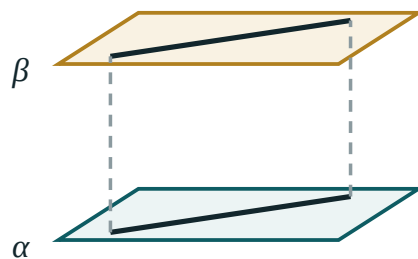
If **two intersecting** lines of one plane are respectively parallel to two lines of another plane, the planes are parallel.

TWO INTERSECTING LINES

One is not enough, and two parallel ones are not enough. A plane containing a single line parallel to β can still cut β — think of a page turning about that line. The two directions must be genuinely different to pin the plane down.

Proof sketch. If the planes met in a line c , then c would lie in α and be parallel to both of the given lines of α — impossible, since they intersect and a line cannot be parallel to two different directions in the same plane. ■

Four properties



$\alpha \parallel \beta$ — a third plane cuts them in parallel lines

A third plane cuts two parallel planes. The two lines of intersection are parallel.

1. A third plane cutting two parallel planes cuts them in **parallel lines**.
2. Segments of parallel lines cut off between two parallel planes are **equal**.
3. Through a point off a plane there is **exactly one** plane parallel to it.
4. If $\alpha \parallel \beta$ and $\beta \parallel \gamma$ then $\alpha \parallel \gamma$ — parallelism of planes is transitive.

Property 2 is the workhorse: it is why the lateral edges of a prism are all equal, and why the cross-sections of a prism parallel to the base are all congruent.

Applications

The prism. Its two bases lie in parallel planes. Property 1 makes corresponding base edges parallel; property 2 makes the lateral edges equal. Both facts are usually assumed — they are theorems, and this is their proof.

The midline of a tetrahedron. A plane through the midpoints of three edges is parallel to a face, because two of its intersecting lines are midlines, each parallel to an edge of that face.

Distance. Between two parallel planes the distance is the same everywhere — the length of any common perpendicular segment.

02 Worked examples

EXAMPLE 1 Prove that in a cube the planes $ABCD$ and $A_1B_1C_1D_1$ are parallel.

- ① $AB \parallel A_1B_1$ and $AD \parallel A_1D_1$.
Opposite sides of square faces.

- ② AB and AD intersect at A .
Two different directions.

- ③ Test applies.

ANSWER

$$ABCD \parallel A_1B_1C_1D_1$$

EXAMPLE 2 Two parallel planes are cut by a third in lines a and b . If a makes 35° with the line of intersection of the cutting plane and a fourth plane, what does b make?

- ① $a \parallel b$
Property 1.
- ② Angles with respectively parallel sides.
- ③ Equal or supplementary; here equal.

ANSWER

$$35^\circ$$

EXAMPLE 3 A tetrahedron $ABCD$. M , N , P are the midpoints of AB , AC , AD . Prove $MNP \parallel BCD$.

① MN is a midline of $\triangle ABC$, so $MN \parallel BC$.

② MP is a midline of $\triangle ABD$, so $MP \parallel BD$.

③ MN and MP intersect at M .

④ BC and BD lie in BCD .

Test applies.

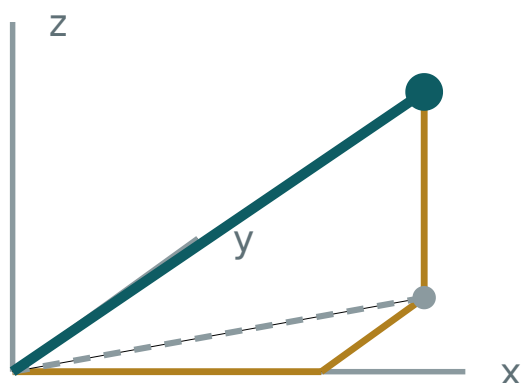
ANSWER

$MNP \parallel BCD$

03 Interactive model

Two sheets of card held parallel, cut by a third — the two cut lines are visibly parallel.

Two planes



x 3

 y^2

z 2

OM 4.12

base diagonal **3.61**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Parallel planes	Parallel tekisliklar	Параллельные плоскости
Coincident planes	Ustma-ust tushuvchi tekisliklar	Совпадающие плоскости
Test for parallel planes	Tekisliklar parallelligi alomati	Признак параллельности плоскостей
Intersecting planes	Kesishuvchi tekisliklar	Пересекающиеся плоскости
Line of intersection	Kesishish chizig'i	Линия пересечения
Distance between planes	Tekisliklar orasidagi masofa	Расстояние между плоскостями
Equal segments	Teng kesmalar	Равные отрезки
Transversal plane	Kesuvchi tekislik	Секущая плоскость

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Two distinct planes can share:

a point only

a line

two points only

nothing but a point

The test for parallel planes needs:

one parallel line

two intersecting parallel lines

three points

a perpendicular

A third plane cuts two parallel planes in:

perpendicular lines

parallel lines

one line

points

Segments of parallel lines between parallel planes are:

equal

proportional

perpendicular

unrelated

Through a point off a plane the number of parallel planes is:

0

1

2

infinitely many

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Name the three positions of two planes

ANSWER parallel, intersecting, coincident

2. Can two planes meet in exactly one point?

ANSWER no

3. State the test for parallel planes

ANSWER Two intersecting lines of one parallel to two of the other

4. Are $ABCD$ and $A_1B_1C_1D_1$ parallel in a cube?

ANSWER yes

5. Are $ABCD$ and ABB_1A_1 parallel?

ANSWER no — they meet in AB

6. How many planes through a point are parallel to a given plane?

ANSWER 1

7. Is parallelism of planes transitive?

ANSWER yes

Medium · the standard question

7 PROBLEMS

1. Prove $ABCD \parallel A_1B_1C_1D_1$ in a cube

ANSWER $AB \parallel A_1B_1$ and $AD \parallel A_1D_1$, intersecting

2. Name a pair of parallel faces of a hexagonal prism

ANSWER the two hexagonal bases

3. Why is one parallel line not enough for the test?

ANSWER The plane can still turn about it

4. Distance between the base and the top of a cube of side 9

ANSWER 9

5. Tetrahedron: prove the midpoint plane $MNP \parallel BCD$

ANSWER Two midlines parallel to two edges of BCD

6. Two parallel planes cut by a third: the lines are

ANSWER parallel

7. Why are the lateral edges of a prism equal?

ANSWER Parallel segments between parallel planes

Hard · stretch and reason

7 PROBLEMS

1. Prove the test for parallel planes

ANSWER A common line would be parallel to two intersecting directions

2. Prove property 2 — segments between parallel planes are equal

ANSWER They form a parallelogram

3. Prove that if $\alpha \parallel \beta$ and a line meets α , it meets β

ANSWER Otherwise it would be parallel to both

4. A plane cuts a cube parallel to a face. Describe the section

ANSWER A square congruent to the face

5. A plane cuts a tetrahedron parallel to a face. Describe the section

ANSWER A triangle similar to that face

6. Prove: three parallel planes cut two transversals in proportional segments

ANSWER Reduce to the plane through the two transversals

7. Give two planes each parallel to the same line that are not parallel to each other

ANSWER Two walls of a room and a vertical line

07 Homework

Homework — 5 tasks

The test needs two *intersecting* lines. Say so in every proof that uses it.

1. Prove that in a cube the planes ABB_1A_1 and DCC_1D_1 are parallel.
2. Tetrahedron $ABCD$: prove the plane through the midpoints of AB , AC , AD is parallel to BCD .
3. Explain in three sentences why one parallel line is not enough for the test.
4. Prove that segments of parallel lines cut off between two parallel planes are equal.
5. Give an example of two planes parallel to the same line that intersect each other.

Parallel projection in space

How a three-dimensional figure is drawn on a flat page — what survives the projection and what does not.

LESSON 27–28

2 × 40 MIN

GEOMETRY 10, §13

IGCSE E14.X

LESSON CLOCK

10'

22'

22'

24'

16'

6'

The construction	10 min
What is preserved	22 min
What is not	22 min
Drawing solids correctly	24 min
Practice	16 min
Homework	6 min

LEARNING OBJECTIVES

- Construct the parallel projection of a point, segment and figure onto a plane.
- State the properties preserved by parallel projection.
- Explain why a circle projects to an ellipse and a square to a parallelogram.
- Draw a correct picture of a cube, prism and pyramid.

TEXTBOOK

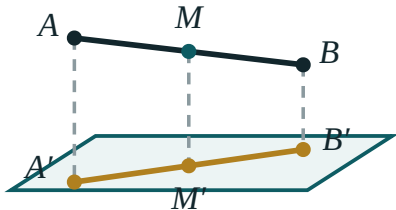
National	pp. 125–136
Cambridge	Core & Extended, pp. 210–216
Workbook	Exercise 13.1

01

Explanation

The construction

Fix a plane α and a direction ℓ not parallel to it. Through each point A of the figure draw the line through A parallel to ℓ ; where it meets α is the **projection** A' .



the midpoint projects to the midpoint

Every point drops along the same direction. The midpoint lands on the midpoint.

WHY IT MATTERS

Every picture of a solid on paper — every diagram in this book — is a parallel projection. Knowing what it preserves is knowing what you may legitimately read off a drawing.

What is preserved

PROPERTY	PRESERVED?
a straight line projects to a straight line (or a point)	yes
parallel lines project to parallel lines	yes
the ratio in which a point divides a segment	yes
the midpoint of a segment	yes
the ratio of lengths on parallel lines	yes
lengths	no
angles	no
the ratio of lengths on non-parallel lines	no

NEVER MEASURE A DRAWING

A right angle in the solid may appear as 60° on the page, and two equal edges may appear to differ. A drawing is a guide to structure, never to size.

What the images look like

FIGURE	ITS PARALLEL PROJECTION
square, rectangle, rhombus	a parallelogram (any one)
parallelogram	a parallelogram
triangle	a triangle (any one)
circle	an ellipse
trapezium	a trapezium — the parallel pair stays parallel

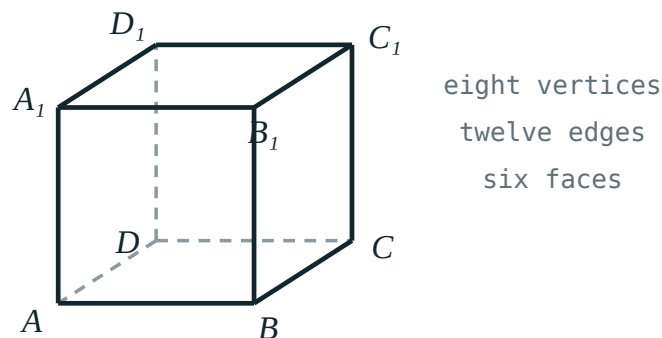
Because **any** triangle can be the image of **any** triangle, and any parallelogram the image of any square, a picture of a cube may be drawn with the base as any convenient parallelogram. That freedom is what makes cabinet projection — the drawing convention of this book — legitimate.

Drawing solids correctly

THE RULES OF A CORRECT PICTURE

1 Parallel edges are drawn parallel. **2** Midpoints are drawn at midpoints. **3** Hidden edges are dashed. **4** The base is drawn as a parallelogram, not as a rectangle. **5** Vertical edges stay vertical.

The standard convention here is **cabinet projection**: the depth axis is drawn at 45° and at half scale, so a unit cube appears with a receding edge of length 0.5 . It is not what the eye sees, but it makes every parallel edge parallel on the page, which is what a geometry diagram needs.



A correct cube: parallel edges parallel, hidden edges dashed, base a parallelogram.

02 Worked examples**EXAMPLE 1** *M* is the midpoint of *AB*. Where is *M'* on *A'B'*?

- ① Parallel projection preserves the ratio of division.
- ② $AM : MB = 1 : 1$
- ③ So $A'M' : M'B' = 1 : 1$.

ANSWERAt the midpoint of *A'B'***EXAMPLE 2** A square is projected. Can its image be a rhombus that is not a square?

- ① A square is a parallelogram.
- ② Any parallelogram can be the image of a square.
- ③ A rhombus is a parallelogram.

ANSWER

Yes

EXAMPLE 3 Can the projection of a triangle be a segment?

① If the plane of the triangle is parallel to the direction of projection...

② ...every point drops onto one line.

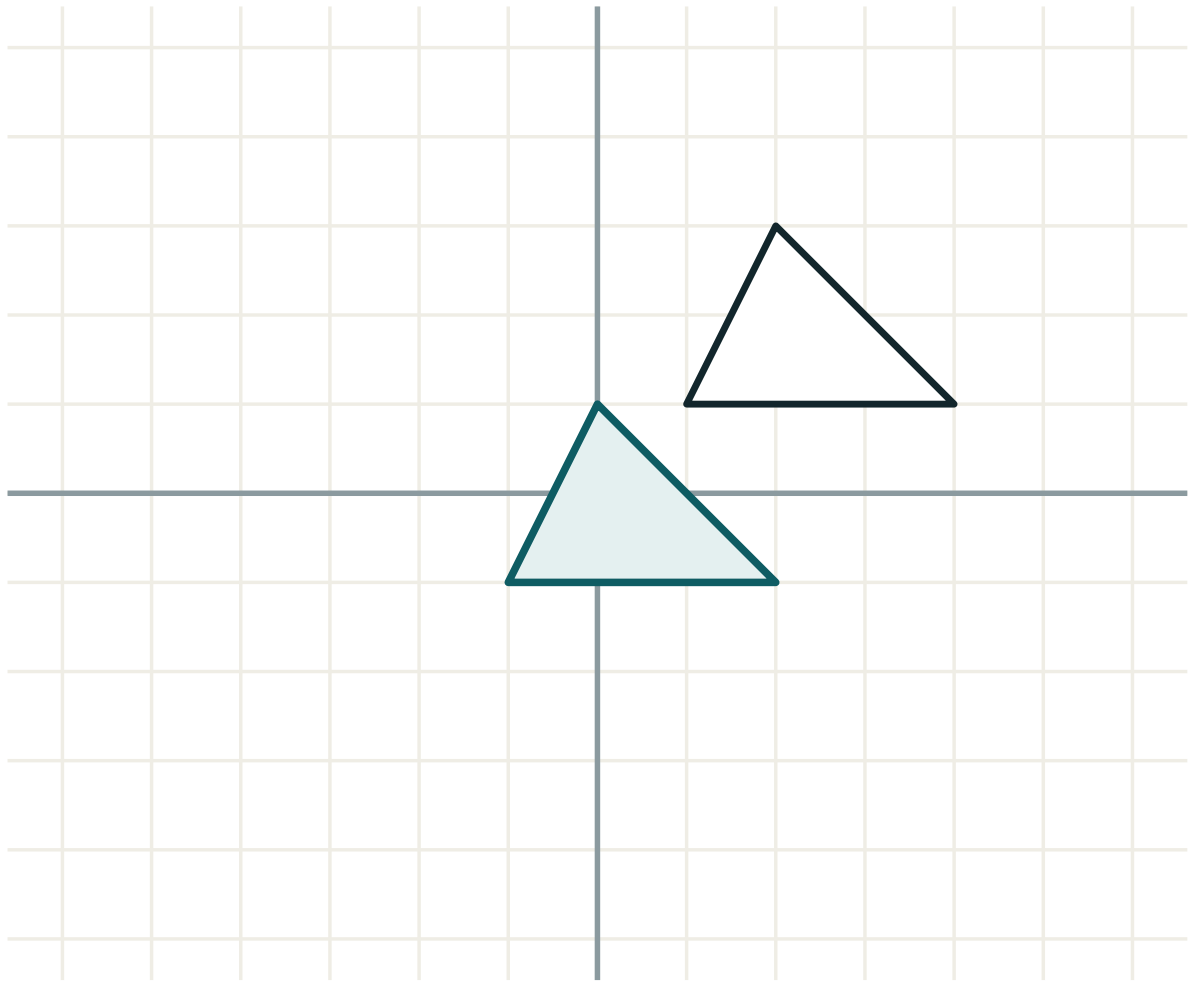
ANSWER

Yes — when the triangle's plane contains the direction of projection

03 **Interactive model**

Hold a wire square in sunlight and turn it — the shadow runs through every parallelogram.

What a projection preserves



transformation **translation**

lengths **unchanged**

angles **unchanged**

orientation **kept**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Parallel projection	Parallel proyeksiya	Параллельное проектирование
Direction of projection	Proyeksiyalash yo'nalishi	Направление проектирования
Projecting line	Proyeksiyalovchi chiziq	Проектирующая прямая
Image (projection)	Proyeksiya	Проекция
Plane of projection	Proyeksiya tekisligi	Плоскость проекции
Preserved property	Saqlanuvchi xossa	Сохраняющееся свойство
Ratio of division	Bo'linish nisbati	Отношение деления
Visible and hidden edges	Ko'rinadigan va ko'rinmaydigan qirralar	Видимые и невидимые рёбра
Correct picture	To'g'ri tasvir	Правильное изображение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Parallel projection preserves:

lengths

angles

the midpoint

areas

The image of a circle is:

a circle

an ellipse

a parabola

a line

The image of a square can be:

only a square

any parallelogram

any quadrilateral

only a rhombus

Angles under parallel projection are:

preserved

doubled

not preserved

halved

Hidden edges in a correct picture are:

omitted

dashed

thickened

coloured

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Image of a straight line

ANSWER a line (or a point)

2. Image of parallel lines

ANSWER parallel lines

3. Image of a midpoint

ANSWER the midpoint

4. Image of a circle

ANSWER an ellipse

5. Image of a square

ANSWER a parallelogram

6. Are lengths preserved?

ANSWER no

7. Are angles preserved?

ANSWER no

Medium · the standard question

7 PROBLEMS

1. Can a square project to a rhombus?

ANSWER yes

2. Can a rectangle project to a trapezium?

ANSWER no — parallel stays parallel

3. Can a triangle project to a segment?

ANSWER yes

4. Is the ratio $AM : MB$ preserved?

ANSWER yes

5. Is the ratio of two perpendicular edges preserved?

ANSWER no

6. In cabinet projection the depth axis is drawn at

ANSWER 45° and half scale

7. Why is the base of a cube drawn as a parallelogram?

ANSWER A square projects to a parallelogram

Hard · stretch and reason

7 PROBLEMS

1. Prove that parallel projection preserves the midpoint

ANSWER The projecting lines cut two parallels proportionally

2. Prove that parallel lines project to parallel lines

ANSWER Their projecting planes are parallel

3. Can two perpendicular lines project to parallel lines?

ANSWER Yes — if their common plane contains the direction

4. A cube of side 1 in cabinet projection: length of the receding edge on paper

ANSWER 0.5

5. Why can any triangle be the image of any triangle?

ANSWER An affine map takes any triangle to any other

6. Which is preserved: the ratio of areas, or the ratio of lengths on non-parallel lines?

ANSWER the ratio of areas

7. Explain why a drawing must never be measured

ANSWER Lengths and angles are not preserved

07 Homework

Homework — 5 tasks

Task 5 must be drawn, with hidden edges dashed.

1. List four properties preserved by parallel projection and three that are not.
2. Explain why a square can project to any parallelogram.
3. Can the projection of a rectangle be a trapezium? Justify your answer.
4. A point divides a segment in the ratio 2 : 3. What is the ratio on the image?
5. Draw a correct picture of a cube and of a hexagonal prism, with hidden edges dashed.

The area of a triangle, and trigonometry in three dimensions

Cambridge insert: one area formula for every triangle, applied inside solids.

LESSON 29–30

2 × 40 MIN

GEOMETRY 10, §13 (EXTENSION)

IGCSE E6.5 · E6.6

LESSON CLOCK

10'

16'

26'

24'

20'

4

The area formula	10 min
Heron	16 min
Sections of solids	26 min
Mixed 3-D problems	24 min
Practice	20 min
Homework	4 min

LEARNING OBJECTIVES

- Use $S = \frac{1}{2}ab \sin C$ for any triangle.
- Use Heron's formula when three sides are known.
- Find the area of a triangular section of a solid.
- Combine the area formula with the sine and cosine rules in a 3-D problem.

TEXTBOOK

National	pp. 137–142
Cambridge	Core & Extended, pp. 300–312
Workbook	IGCSE Exercise 6.5

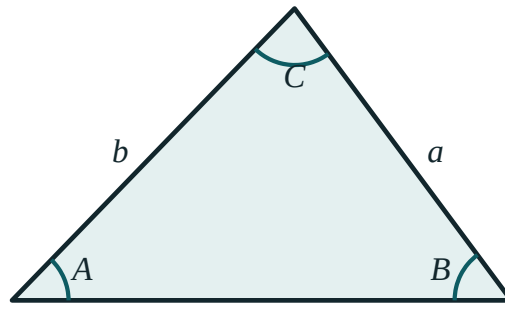
01

Explanation

One formula for every triangle

$$S = \frac{1}{2}ab \cdot \sin C$$

Two sides and the angle **between them**. It reduces to $\frac{1}{2}bh$ when $C = 90^\circ$, because $\sin 90^\circ = 1$ — the familiar formula is the special case.



$$a / \sin A = b / \sin B = c / \sin C$$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

The labelling convention: side a opposite angle A. The area uses the angle between the two named sides.

THE ANGLE MUST BE THE INCLUDED ONE

$\frac{1}{2}ab \cdot \sin A$ is wrong: A is not between a and b. Name the two sides first, then take the angle they enclose.

Heron's formula

When all three sides are known and no angle is:

$$S = \sqrt{s(s-a)(s-b)(s-c)}, \text{ where } s = \frac{a+b+c}{2}$$

For 5, 6, 7: $s = 9$ and $S = \sqrt{9 \cdot 4 \cdot 3 \cdot 2} = \sqrt{216} \approx 14.7$.

WHICH TO USE

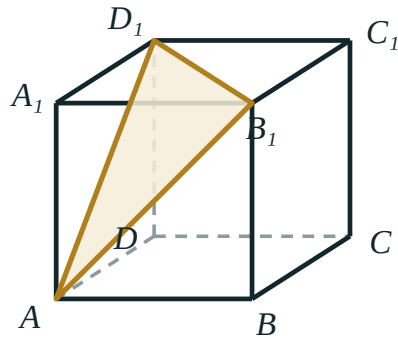
Two sides and the included angle → the sine formula. Three sides → Heron, or find an angle by the cosine rule and then use the sine formula. Both give the same number.

Areas of sections

The sections of Lesson 12–13 are polygons in space; their areas are computed by splitting into triangles and applying the formula to each.

Example. Cube of side 6. The section through A, C, B_1 is an equilateral triangle of side $6\sqrt{2}$. Using the formula with $C = 60^\circ$:

$$S = \frac{1}{2} \cdot 72 \cdot \sin 60^\circ = 36 \cdot \frac{\sqrt{3}}{2} = 18\sqrt{3} \approx 31.2$$



the section
A B₁ D₁ is
an equilateral
triangle

The triangular section, and the triangle drawn flat beside it.

Mixed problems

A full 3-D problem chains the three tools:

1. Find the lengths of the section's sides by Pythagoras in the solid.
2. Find an angle by the cosine rule in the extracted triangle.
3. Find the area by $\frac{1}{2}ab \sin C$.

Example. Cuboid $4 \times 6 \times 12$. Section through $A(0,0,0)$, $B(4,0,0)$ and $C_1(4,6,12)$: $AB = 4$, $BC_1 = \sqrt{36 + 144} \approx 13.42$, and $AB \perp BC_1$ because AB is perpendicular to the whole face BCC_1B_1 . So the area is $\frac{1}{2} \cdot 4 \cdot 13.42 \approx 26.8$.

02 Worked examples

EXAMPLE 1 Find the area of a triangle with $a = 7$, $b = 9$, $C = 40^\circ$.

$$① \quad S = \frac{1}{2} \cdot 7 \cdot 9 \cdot \sin 40^\circ$$

$$② \quad = 31.5 \times 0.6428$$

$$③ \quad \approx 20.2$$

ANSWER

≈ 20.2 square units

EXAMPLE 2 Find the area of a triangle with sides 8, 11 and 13.

$$① \quad s = 16$$

$$② \quad S = \sqrt{16 \cdot 8 \cdot 5 \cdot 3}$$

$$③ \quad = \sqrt{1920} \approx 43.8$$

ANSWER

≈ 43.8

EXAMPLE 3 Cube of side 6. Find the area of the section through A, C, B_1 .

- ① All three sides are face diagonals, $6\sqrt{2}$.
Equilateral.

② $S = \frac{1}{2} \cdot 72 \cdot \sin 60^\circ$

③ $= 18\sqrt{3} \approx 31.2$

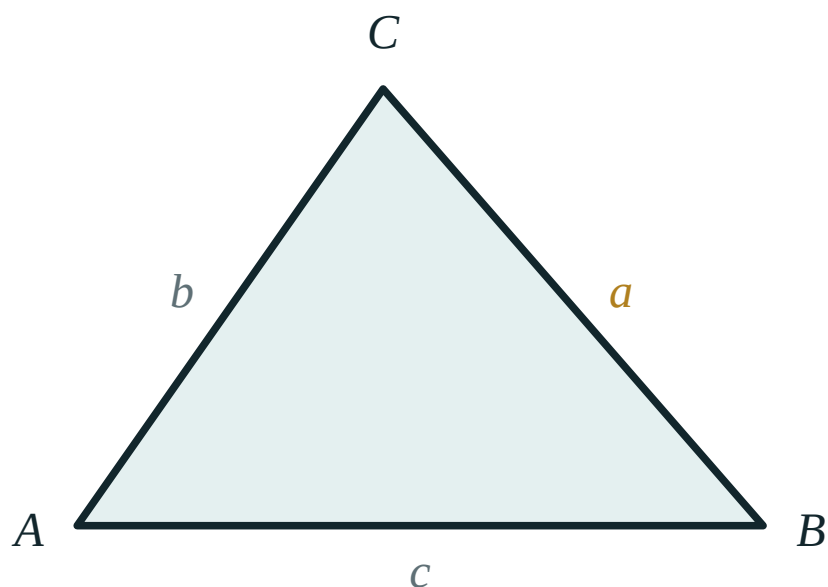
ANSWER

$$18\sqrt{3} \approx 31.2$$

03 **Interactive model**

Extract the triangle from the solid and draw it flat before any calculation.

Two sides and the included angle



b 7

c 9

A 55°

a 7.6

B 49°

C 76°

area 25.8

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Area of a triangle	Uchburchak yuzasi	Площадь треугольника
Included angle	Ichki burchak	Угол между сторонами
Heron's formula	Geron formulasi	Формула Герона
Semi-perimeter	Yarim perimetr	Полупериметр
Triangular section	Uchburchakli kesim	Треугольное сечение
Sine rule	Sinuslar teoremasi	Теорема синусов
Cosine rule	Kosinuslar teoremasi	Теорема косинусов
Three-dimensional problem	Fazoviy masala	Пространственная задача
Extracted triangle	Ajratilgan uchburchak	Выделенный треугольник

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The area formula $\frac{1}{2} ab \sin C$ needs:

any angle

the angle between a and b

the largest angle

a right angle

Heron's formula needs:

two sides and an angle

three sides

two angles

the height

s in Heron's formula is:

the perimeter

half the perimeter

the longest side

the area

Area of a triangle with $a = 6$, $b = 8$, $C = 30^\circ$:

12

24

48

6

The section of a cube through A , C , B_1 is:

right-angled

equilateral

isosceles only

scalene

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Area with $a = 4$, $b = 5$, $C = 90^\circ$

ANSWER 10

2. Area with $a = 6$, $b = 8$, $C = 30^\circ$

ANSWER 12

3. Area with $a = 10$, $b = 10$, $C = 60^\circ$

ANSWER $25\sqrt{3}$

4. Heron: sides 3, 4, 5

ANSWER 6

5. Heron: sides 5, 6, 7

ANSWER $\sqrt{216} \approx 14.7$

6. s for sides 8, 11, 13

ANSWER 16

7. Area with $a = 12$, $b = 5$, $C = 90^\circ$

ANSWER 30

Medium · the standard question

7 PROBLEMS

1. Area with $a = 7$, $b = 9$, $C = 40^\circ$

ANSWER ≈ 20.2

2. Heron: sides 8, 11, 13

ANSWER ≈ 43.8

3. Cube side 6: area of section ACB_1

ANSWER $18\sqrt{3}$

4. Cube side 4: area of the diagonal section ABC_1D_1

ANSWER $16\sqrt{2}$

5. Triangle 9, 12, 15: area

ANSWER 54

6. Two sides 10 and 14 enclose 120° . Area

ANSWER $35\sqrt{3} \approx 60.6$

7. Cuboid $4 \times 6 \times 12$: area of section ABC_1

ANSWER ≈ 26.8

Hard · stretch and reason

7 PROBLEMS

1. Cube side a : area of the regular hexagonal section

ANSWER $\frac{3\sqrt{3}}{4}a^2$

2. Square pyramid, base 10, height 12: area of one lateral face

ANSWER 65

3. Same pyramid: total lateral surface area

ANSWER 260

4. Triangle with sides 13, 14, 15: area and the height on the 14-side

ANSWER 84, height 12

5. Find the angle when $a = 8$, $b = 10$ and the area is 20

ANSWER 30° or 150°

6. Regular tetrahedron edge 6: total surface area

ANSWER $36\sqrt{3}$

7. Prove $S = \frac{1}{2}ab \sin C$ from $\frac{1}{2}bh$

ANSWER $h = a \sin C$

07 Homework

Homework — 5 tasks

Draw the extracted triangle for every 3-D question.

- Find the area of a triangle with $a = 11$, $b = 6$, $C = 55^\circ$.
- Find the area of a triangle with sides 7, 9 and 12 by Heron's formula.
- Cube of side 8: find the area of the section through A , C , B_1 .
- Square pyramid, base 12, height 8: find the area of one lateral face and the total lateral area.

5. Two sides 9 and 12 enclose an angle whose sine is 0.6. Find the area and both possible angles.

Control work 2, and the quarter review

Parallelism in space — lines, planes, projection and area — in one paper, then the map.

LESSON 31–32

2 × 40 MIN

GEOMETRY 10, NAZORAT ISHI 2

IGCSE E4 REVIEW

LESSON CLOCK

3

40'

12'

22'

10'

3

Instructions	3 min
The paper	40 min
Answers	12 min
Rewrite	22 min
Concept map	10 min
Targets	3 min

LEARNING OBJECTIVES

- Apply the whole quarter in one assessment.
- Quote a test or a theorem by name in every proof.
- Classify each lost mark and rewrite the solution.
- Build a concept map of parallelism in space.

TEXTBOOK

National	pp. 143–146
Cambridge	Core & Extended, pp. 313–316
Workbook	Control paper G2

01

Explanation

The paper — 25 marks, 40 minutes

Q	TASK	MARKS	FROM
1	List every edge of a cube skew to CD , and name one common perpendicular	4	L21–22
2	Prove $B_1C_1 \parallel ABCD$, quoting the test	4	L23–24
3	Prove $ABB_1A_1 \parallel DCC_1D_1$, quoting the test	4	L25–26
4	Give a counterexample showing perpendicularity in space is not transitive	3	L19–20
5	List four properties preserved by parallel projection and two that are not	4	L27–28
6	Cube of side 6: find the area of the section through A, C, B_1	6	L29–30

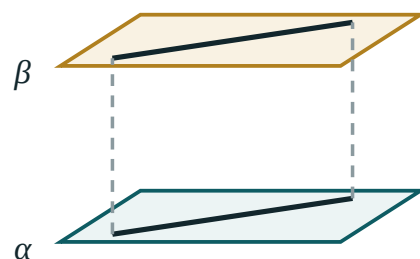
PROOFS ARE MARKED ON THE REASON

Q2 and Q3 each give 3 of their 4 marks for naming the test and verifying its conditions. A correct conclusion with no reason scores one.

The concept map

Six boxes, links as sentences:

- **two lines** → **three cases** — “parallel, meeting, or skew”
- **skew** → **common perpendicular** — “exactly one, and it is the distance”
- **line and plane** → **the test** — “parallel to *some* line of the plane”
- **two planes** → **the test** — “two *intersecting* lines, respectively parallel”
- **parallel planes** → **equal segments** — “why a prism’s lateral edges are equal”
- **projection** → **what a drawing may be trusted for** — “structure, never size”



$\alpha \parallel \beta$ – a third plane cuts them in parallel lines

The picture behind two of the six boxes.

Looking forward

Quarter III is perpendicularity — the theorem of the three perpendiculars, the distance from a point to a plane, and the dihedral angle. Every proof will use a parallelism result from this quarter, and every construction the projection rules of Lesson 27–28.

ONE HABIT TO CARRY FORWARD

Name the test, then verify each of its conditions in turn. Two of the four marks in every proof question this quarter were for exactly that.

02 Worked examples

EXAMPLE 1 Model answer, Q2: prove $B_1C_1 \parallel ABCD$.

- 1 $B_1C_1 \not\subset ABCD$
Condition 1 of the test.
- 2 $B_1C_1 \parallel BC$
Opposite sides of square BCC_1B_1 .
- 3 $BC \subset ABCD$
Condition 2.
- 4 By the test for a line parallel to a plane.

ANSWER

$$B_1C_1 \parallel ABCD$$

EXAMPLE 2 Model answer, Q3: prove $ABB_1A_1 \parallel DCC_1D_1$.

- 1 $AB \parallel DC$ and $AA_1 \parallel DD_1$.
- 2 AB and AA_1 meet at A .
Two intersecting lines.
- 3 DC and DD_1 lie in DCC_1D_1 .
- 4 By the test for parallel planes.

ANSWER

$$ABB_1A_1 \parallel DCC_1D_1$$

EXAMPLE 3 Model answer, Q6: cube of side 6, section through A, C, B₁.

- ① Each side is a face diagonal, $6\sqrt{2}$.

By rule R1 three times.

- ② Equilateral, so every angle is 60° .

- ③ $S = \frac{1}{2} \cdot 72 \cdot \sin 60^\circ$

- ④ $= 18\sqrt{3} \approx 31.2$

ANSWER

$$18\sqrt{3} \approx 31.2$$

03 Interactive model

Work Q2 and Q3 on the board, writing the test out in full before applying it.

The quarter in ten questions

Parallel lines in space must be:

non-meeting

coplanar and non-meeting

perpendicular

equal

Perpendicularity in space is:

transitive

not transitive

symmetric only

undefined

How many edges of a cube are skew to a given edge?

2

3

4

6

Two skew lines have how many common perpendiculars?

0

1

2

∞

The test for $\ell \parallel \alpha$ needs:

every line of α

one line of α

a perpendicular

two planes

The test for parallel planes needs:

one parallel line

two intersecting parallel lines

three points

a common point

A third plane cuts two parallel planes in:

parallel lines

perpendicular lines

a point

nothing

Parallel projection preserves:

lengths

angles

midpoints

areas

The image of a circle is:

a circle

an ellipse

a line

a parabola

Cube side 6: area of the section ACB_1 :

18

$18\sqrt{3}$

36

$36\sqrt{2}$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Test (criterion)	Alomat	Признак
Justification	Asoslash	Обоснование
Concept map	Tushunchalar xaritasi	Карта понятий
Revision	Takrorlash	Повторение
Self-assessment	O'z-o'zini baholash	Самооценка
Target	Maqsad	Цель
Counterexample	Qarshi misol	Контрпример

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A proof scores most of its marks for:

the conclusion

naming and checking the test

neatness

speed

The test for parallel planes fails with:

two intersecting lines

one line only

three lines

a perpendicular

A drawing may be trusted for:

lengths

angles

structure

areas

Quarter III begins with:

- volume
- perpendicularity in space
- vectors
- probability

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Edges skew to CD in a cube

ANSWER $AA_1, BB_1, A_1D_1, B_1C_1$

2. Is $B_1C_1 \parallel ABCD$?

ANSWER yes

3. Is $ABB_1A_1 \parallel DCC_1D_1$?

ANSWER yes

4. Is perpendicularity transitive in space?

ANSWER no

5. Does projection preserve midpoints?

ANSWER yes

6. Does projection preserve angles?

ANSWER no

7. Area with $a = 6, b = 8, C = 30^\circ$

ANSWER 12

Medium · the standard question

7 PROBLEMS

1. Prove $B_1C_1 \parallel ABCD$

ANSWER Test: $B_1C_1 \parallel BC \subset ABCD$

2. Prove $ABB_1A_1 \parallel DCC_1D_1$

ANSWER Two intersecting parallel pairs

3. Common perpendicular of CD and AA_1

ANSWER AD

4. Cube side 6: area of section ACB_1

ANSWER $18\sqrt{3}$

5. Cube side 6: distance between AB and CC_1

ANSWER 6

6. Image of a square under projection

ANSWER a parallelogram

7. Angle between AB and DC_1 in a cube

ANSWER 45°

Hard · stretch and reason

7 PROBLEMS

1. Cube side a : distance between AB and B_1D_1

ANSWER $\frac{a}{\sqrt{2}}$

2. Prove segments of parallel lines between parallel planes are equal

ANSWER They bound a parallelogram

3. Cube side 6: area of the midpoint hexagonal section

ANSWER $27\sqrt{3} \approx 46.8$

4. Prove: if $\ell \parallel \alpha$ and $\ell \parallel \beta$ with $\alpha \cap \beta = m$ then $\ell \parallel m$

ANSWER Converse theorem in each plane

5. Tetrahedron: prove the midpoint plane is parallel to a face

ANSWER Two midlines, two directions

6. Give two planes parallel to the same line that intersect

ANSWER Two walls and a vertical line

7. Heron: sides 8, 11, 13

ANSWER ≈ 43.8

07 Homework

Homework — 4 tasks

Bring the concept map to the first lesson of Quarter III.

1. Rewrite in full every control-work question that lost a mark, naming the test used.
2. Finish the concept map with all six links written as sentences.
3. Cube of side 10: find the area of the section through A, C, B_1 and the distance between AB and CC_1 .

4. Write your target for Quarter III in one checkable sentence, and date it.